

Worksheet 5: Section 1.5

Problem Solving and Quadratic Applications

Name: _____

Date: _____

Problem-Solving Strategy

6-Step Problem-Solving Strategy:

1. **Read** – Understand the problem. What are you solving for?
2. **Define** – Assign variables. Figure out how they relate.
3. **Draw** – If possible, draw a picture or graph to visualize.
4. **Equation** – Write an equation using the given information.
5. **Solve** – Solve the equation algebraically (using factoring, quadratic formula, etc.).
6. **Check** – Verify your answer works in the original problem.

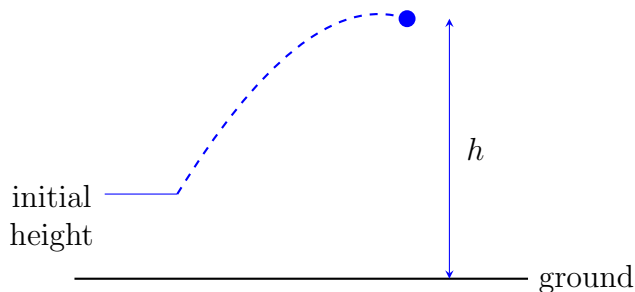
Practice Problems

Show all work. Box your final answers.

Problem 1. A ball is thrown from an initial height of 7 feet with an initial upward velocity of 21 ft/s. The ball's height h (in feet) after t seconds is given by:

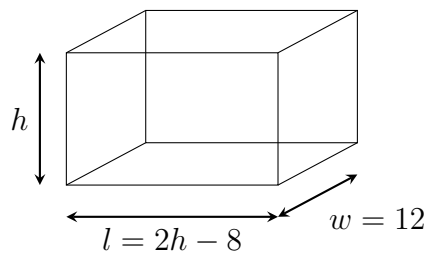
$$h = 7 + 21t - 16t^2$$

Find all values of t for which the ball's height is 13 feet. Round your answer(s) to the nearest hundredth.



Solution:

Problem 2. Dmitri is having a fish tank constructed to fit in a niche in his wall. The width must be 12 inches, while the length is twice the height decreased by 8 inches. The total volume of the tank must be 6840 cubic inches. What are the dimensions of the tank?



Solution:

Answer Key

Answer Key

Problem 1: $t \approx 0.42$ seconds or $t \approx 0.89$ seconds

Problem 2: Width = 12 inches, Length = 30 inches, Height = 19 inches